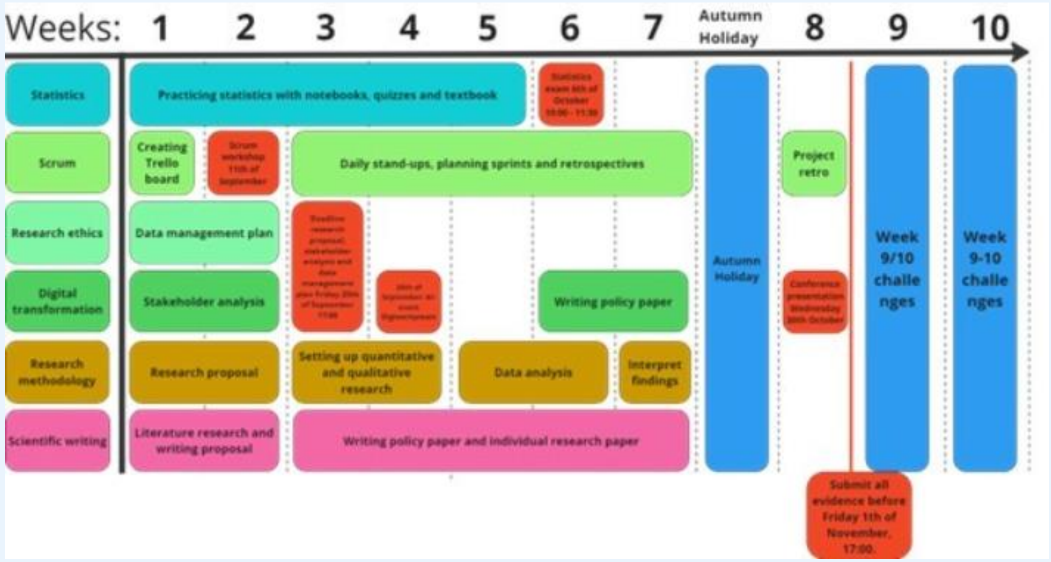
Week 9-10 challenges

ADS&AI

Week 9-10

- Blocks lasts for 8 weeks
- In week 9 and 10 lecturers grade your work and prepare for next block

- We want you to spend this time on developing yourself further



Set-up challenges

- Relevant to the program
- Individual, in duo’s or in teams
- No full-time supervision, but lecturer available for Q&A
- Often in collaboration with industry
- Hackathons
- You can do multiple challenges in one block

- Good for your portfolio and finding a future job!

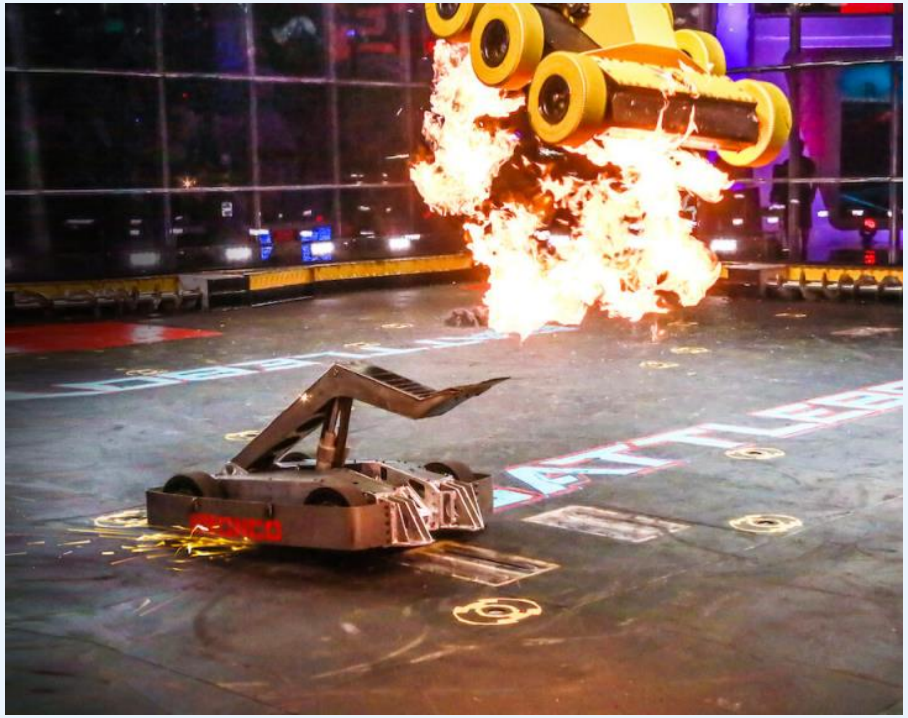
Grading

- Content is not graded, but your participation is!
- The complexity and time-intensity of a challenge define the number of **points** you get for completing the challenge.
- In Block D the **prior requirements** for the professional and personal ILO's have a requirement of **10 points** of challenges.
- Points are awarded after completing a challenge.

Block B Challenges

- Robotics – Battle Bots
- Football Focus: Season Unveiled
- Company challenge: DKG
- Hackathon for Good
- Short story/Film script
- Company challenge: Corporate Game Chick (Part 2)
- Real-time motion tracking with IMU and Blender
- Mathematics
- Open Day
- Mirro
- Zooniverse
- Own challenge

Robotics - Battle Bots



- Build a controller for a balloon popping battle bot.
- PyBullet simulation environment.
- Year 1's should join teams with Year 2's
- Robotics club membership not required

Number of points: 3

Teams 2-3

Student limit: None

Contact person: Jason Harty

Football Focus: Season Unveiled

- EDA on a dummy football season.
- You need to make decisions like finding best players, players to sell etc. based on data.
- Predict future results.

Number of points: 2

Individual

Student limit: 45

Contact person: Karna Rewatkar

Company challenge: DKG

- **Two project proposals to choose from**
- Collaborate with DKG IT employees to explore opportunities where AI and/or Data Science can enhance the department's performance. Identify areas for improvement and propose solutions to optimize workflows or processes.
- When customers order kitchen, their designs must meet specific constraints. For this project, train a model to automatically detect anomalies in a dataset of kitchen orders, ensuring compliance with the constraints.

Number of points: 3

Teams of 4 students

Student limit: 40

Contact person: Margot Neggers

Hackathon for Good

- From 31 January to 2 February (week 9) Hackathons for Good hosts an in-person Hackathon in The Hague.
- In a timespan of 48 hours you will work with a multidisciplinary team on a challenge to accelerate purpose-driven innovation.
- Check the site for more information:
- https://www.hackathonforgood.org/hackathons/the-hague-7

Number of points: 3	Individual	Student limit: None	Contact person: Avril Hayden

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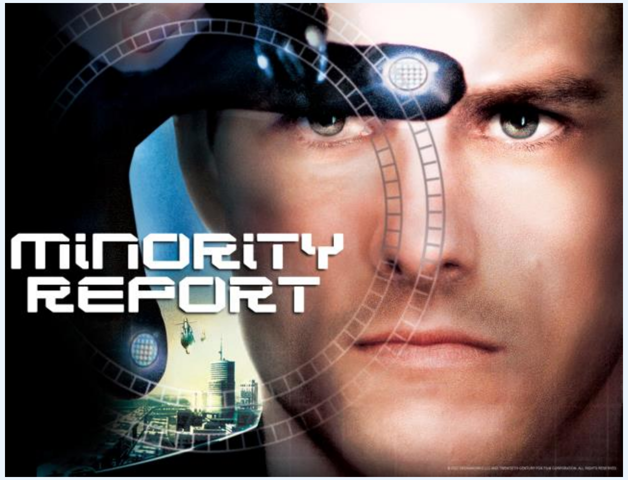
Short Story/Film Script



Imagine a world where the boundaries of technology have been pushed beyond what we know today. The year is 2050, and we challenge you to visualize and share this world with us. Your task is to create a narrative that intertwines the incredible advancements of AI with the ethical challenges it brings:

- Craft a compelling story or film script that paints a vivid picture of the world in 2050.
- Accompany your narrative with a detailed description of the technological advancements in AI that your story of film script highlights. Dive into the ethical implications and challenges of these advancements.
- Your creations should not only be informative but also engaging and fun for the audience.

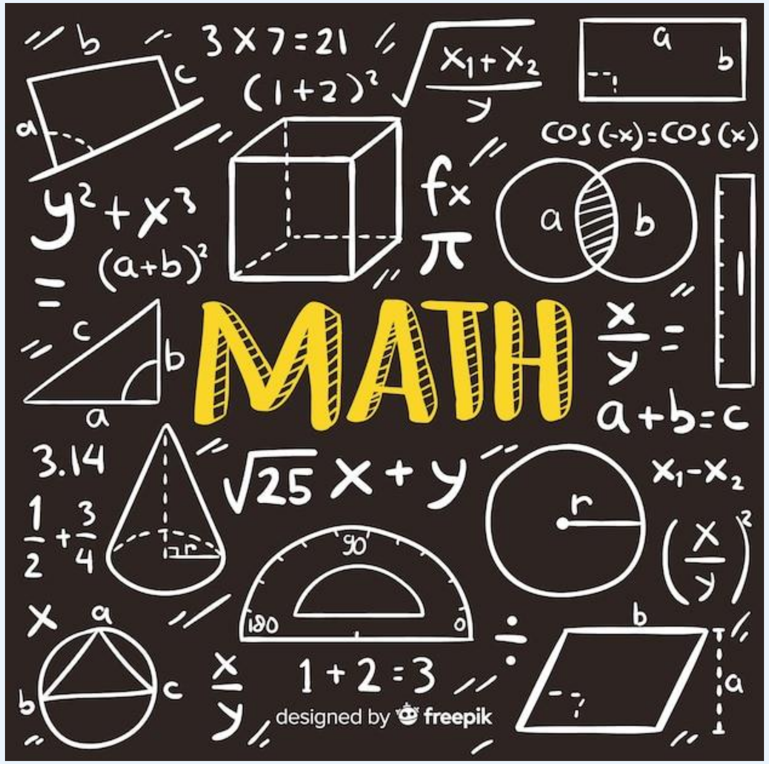
Number of points: 2

Individual/Pairs

Student limit: 30

Contact person: Irene van Blerck / Oscar Bastiaens

Mathematics



- **Struggling with topics on mathematics?**
- In this challenge you can follow some selected mathematics courses that will improve your understanding.
- The courses are chosen from Khan Academy after discussions with the contact person for this challenge.
- The points depend on the number of courses you take and their duration/difficulty.

Number of points: TBD

Individual

Student limit: 10

Contact person: Arash Sadeghzadeh

Company Challenge: Corporate Game Chick (PART 2)

www.corporategamechick.com

- **Problem:** The company is developing a serious game to help employees understand compliance policies. The game will include a <u>Virtual Compliance Assistant</u> (chatbot).
- **Challenge:** Continue the development of the chatbot according to the feedback provided by the company for the first part of the challenge. The new tasks involve:
  - Focusing on interaction and dialog between chatbot and employee (chatbot asking clarification questions).
  - Incorporating user-specific role information.
- The challenge is <u>open to all student</u> (not only the ones that participated in the first part of challenge)

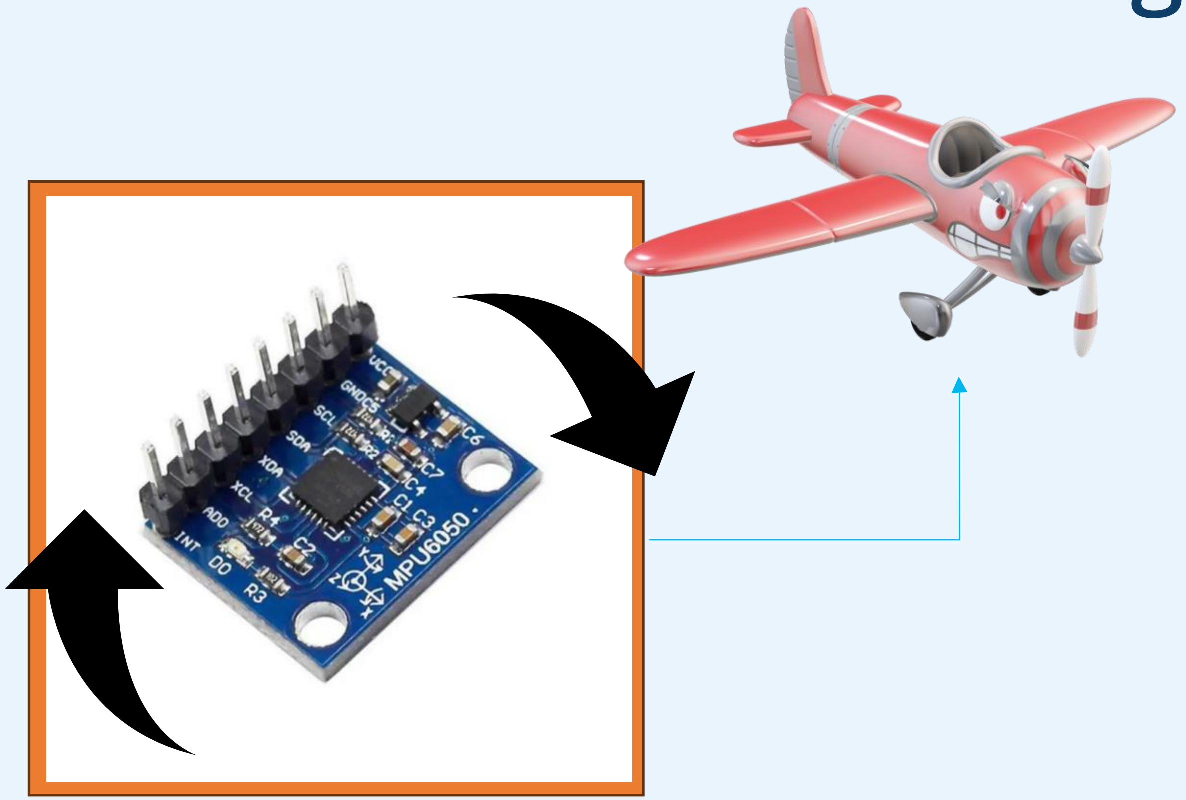
Number of points: 3

Duo’s

Student limit: 16

Contact person: Edirlei Lima

Real-Time Motion Tracking with IMU and Blender



- You will work with an **IMU sensor** and an **Arduino** to capture **motion data** (e.g., orientation and rotation) and **process it to control a 3D model in Blender**.
- The model **must respond in real-time** to the movement and rotation of the IMU, bringing your chosen object to life.
- Creativity is key: avoid using basic shapes like cubes or spheres and **aim for a model/creation with more complexity and character**.

Number of points: 3	Teams of 4 students	Student limit: 24 6 groups max	Contact person: Shival Indermun

Open day challenge

Note: You can only take part in this challenge if you have not done so already!

- Participate in an open day/evening to attract future students
- This block you could be a part of the open day: **Saturday 8 February (in week 10)**
- Tasks: collect, exhibit and communicate your work
- Extra task: test and assist our robot open day assistant
- **Limited availability**: write a short motivation if you want to join

Number of points: 2	Individual	Student limit: 5	Contact person: Margot Neggers

Mirro modules

Note: You can only take part in this challenge if you have not done so already!

- Do you want to know more about topics like anxiety and stress relief?
- In this challenge you can choose a minimum of 3 Mirro modules
- https://account.mirro.nl/api/surfconext/signon/buas
- After completing the modules write a short (max 1 A4) reflection on your main take-aways (without personal information)
- You can do this challenge only once!

Number of points: 1

Individual

Student limit: none

Contact person: Margot Neggers

Breda University OF APPLIED SCIENCES

Zooniverse

Note: You can only take part in this challenge if you have not done so already!

ZOONIVERSE

- People-powered research
- Classify data for different research projects
- Examples:
  - Classify bird calls to help AI powered biodiversity monitoring
  - Count Galapagos Iguanas from aerial photographs to help save them from extinction!
  - Count penguins to understand if penguin populations are in decline.
  - Identify galaxies to help the European Space Agency train AI algorithms

Number of points: 1

Individual

Student limit: none

Contact person: Avril Hayden

Own challenge

- If you did or plan to do your own extra-curricular activity such as
  - (summer) school
  - Hackathon
  - Self-study topic
  - Own coding project
- This might also be eligible to count as a challenge
- However: discuss this with your mentor!

Number of points: TBD

Individual/duo's

Student limit: None

Contact person: Mentor or content expert

Preference

- Use this link to convey your preferences on a challenge:
- https://forms.office.com/e/Qj7AtJp7Cd
- The deadline is **Friday 20th of December!**
- Division is communicated in week 7